

Comment on “Localization of the Maximal Entropy Random Walk”

Abstract: Burda et al. [Phys. Rev. Lett. 102, 160602 (2009)] asserted that the discrete-time probability distribution $\pi_i(t)$ of the Maximal Entropy Random Walk (MERW) converges to the stationary profile $\pi_i^* = \psi_{0,i}^2$ as $t \rightarrow \infty$. This Comment demonstrates that for any connected bipartite graph, the Markov transition matrix possesses a peripheral eigenvalue $\mu = -1$, inducing period-two dynamics. Consequently, although π^* is stationary, the discrete-time distribution fails to converge to π^* from generic initial conditions, approaching a persistent parity oscillation. Therefore, the convergence claim fails for bipartite lattices, including square and ladder graphs, unless parity averaging, aperiodic modifications, or continuous-time formulations are introduced.

I. Introduction

The Maximal Entropy Random Walk (MERW) defines stochastic trajectories via global path entropy maximization. On a finite, connected graph $G = (V, E)$, the discrete-time stationary probability distribution is claimed to relax to:

$$\pi_i^* = \psi_{0,i}^2$$

where ψ_0 is the principal right eigenvector of the adjacency matrix A [1]. Specifically, Ref. [1] asserts: “using spectral properties of the matrix P_{ij} , one can show that $\pi_i(t)$ reaches for $t \rightarrow \infty$ a unique stationary state π_i^* obeying (2)”.

However, discrete-time Markov chain convergence requires aperiodicity. If the transition matrix P has other eigenvalues on the unit circle, the chain fails to mix. This Comment demonstrates that since the square lattice and ladder network in Ref. [1] are bipartite, P has a peripheral eigenvalue $\mu = -1$. Consequently, while π^* remains valid, the instantaneous distribution $\pi(t)$ fails to converge to π^* from generic initial conditions, approaching a persistent two-cycle.

II. Spectral Projection and Similarity Transformation

Let $G = (V, E)$ denote a finite, connected, bipartite graph. The vertex set consists of two disjoint, independent partitions $V = V_1 \cup V_2$. Ordering the vertices conformally yields the symmetric adjacency matrix:

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}$$

where B is a matrix of dimensions $|V_1| \times |V_2|$. Let λ_0 denote the spectral radius of A . Since G is connected, the Perron-Frobenius theorem establishes that λ_0 is a simple, positive eigenvalue, and the corresponding principal right eigenvector

ψ_0 satisfies $\psi_{0,i} > 0$ for all $i \in V$. Conformal partitioning of ψ_0 yields:

$$\psi_0 = \begin{pmatrix} \psi_0^{(1)} \\ \psi_0^{(2)} \end{pmatrix}$$

The eigenvalue equation $A\psi_0 = \lambda_0\psi_0$ implies the system:

$$B\psi_0^{(2)} = \lambda_0\psi_0^{(1)}$$

$$B^T\psi_0^{(1)} = \lambda_0\psi_0^{(2)}$$

Define the spectral test vector ψ_- by:

$$\psi_- = \begin{pmatrix} \psi_0^{(1)} \\ -\psi_0^{(2)} \end{pmatrix}$$

Direct application of A to ψ_- yields:

$$A\psi_- = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} \psi_0^{(1)} \\ -\psi_0^{(2)} \end{pmatrix} = \begin{pmatrix} -B\psi_0^{(2)} \\ B^T\psi_0^{(1)} \end{pmatrix}$$

Substitution of the system equations into this matrix product yields:

$$\begin{pmatrix} -B\psi_0^{(2)} \\ B^T\psi_0^{(1)} \end{pmatrix} = \begin{pmatrix} -\lambda_0\psi_0^{(1)} \\ \lambda_0\psi_0^{(2)} \end{pmatrix} = -\lambda_0\psi_-$$

Therefore, $-\lambda_0$ is a simple eigenvalue of A with associated right eigenvector ψ_- .

Define the transition matrix P for the discrete-time Maximal Entropy Random Walk via the diagonal similarity transformation:

$$P = \frac{1}{\lambda_0} D^{-1} A D$$

where $D = \text{diag}(\psi_0)$. Since A is real and symmetric, A is strictly diagonalizable. Similarity of P to $\lambda_0^{-1}A$ establishes that P is also diagonalizable, Jordan blocks do not occur, and their spectra are linked by the identical relation:

$$\text{Spec}(P) = \frac{1}{\lambda_0} \text{Spec}(A)$$

The presence of $\pm\lambda_0 \in \text{Spec}(A)$ establishes:

$$\{1, -1\} \in \text{Spec}(P)$$

The right eigenvector v_2 of P corresponding to the peripheral eigenvalue $\mu_2 = -1$ is obtained via:

$$v_2 = D^{-1}\psi_- \implies v_{2,i} = \frac{\psi_{-,i}}{\psi_{0,i}} = \begin{cases} 1 & \text{if } i \in V_1 \\ -1 & \text{if } i \in V_2 \end{cases}$$

III. Dual Basis Expansion and Asymptotic Oscillations

Let $\pi(t)$ represent the state probability row vector under iterated applications of P . We apply a spectral expansion to the initial state $\pi(0)$ in the dual basis of P 's left and right eigenvectors $\{u_\alpha^T, v_\alpha\}$, subject to the biorthogonality condition $u_\alpha^T v_\beta = \delta_{\alpha\beta}$.

A. Subspace Biorthogonality Verification

Let the stationary mode ($\mu_1 = 1$) be defined by $u_1^T = \pi^*$ and $v_1 = 1$, where $\pi_i^* = \psi_{0,i}^2$ subject to the normalization $\sum_i \psi_{0,i}^2 = 1$. Let the oscillatory mode ($\mu_2 = -1$) be defined by v_2 and the left eigenvector $u_2^T = w^T$, where $w_i = \psi_{0,i}\psi_{-,i}$ (satisfying $w^T P = -w^T$). Element-wise, this vector expands to:

$$w_i = \begin{cases} \psi_{0,i}^2 & \text{if } i \in V_1 \\ -\psi_{0,i}^2 & \text{if } i \in V_2 \end{cases}$$

Symmetry of A requires orthogonality of eigenvectors corresponding to distinct eigenvalues. The relation $\psi_0 \perp \psi_-$ yields:

$$w^T v_1 = \sum_{i \in V} w_i = \sum_{i \in V_1} \psi_{0,i}^2 - \sum_{i \in V_2} \psi_{0,i}^2 = \psi_0^T \psi_- = 0$$

Therefore, the L_2 mass of the Perron vector is split equally between the two partitions. Normalization and biorthogonality constraints for the dominant subspace are satisfied:

$$u_1^T v_2 = \sum_{i \in V_1} \psi_{0,i}^2 - \sum_{i \in V_2} \psi_{0,i}^2 = 0$$

$$w^T v_2 = \sum_{i \in V} (\psi_{0,i}\psi_{-,i}) \left(\frac{\psi_{-,i}}{\psi_{0,i}} \right) = \sum_{i \in V} \psi_{-,i}^2 = \sum_{i \in V} \psi_{0,i}^2 = 1$$

B. Eigenvalue Decay and Asymptotic Oscillation

The Perron-Frobenius theorem for irreducible non-negative matrices implies that a connected bipartite graph possesses exactly two eigenvalues of maximum modulus, λ_0 and $-\lambda_0$. Remaining eigenvalues are bounded by:

$$|\mu_\alpha| < 1 \quad \text{for all } \alpha \geq 3$$

For any initial distribution $\pi(0)$, the projection onto the oscillatory mode is $c_2 = \pi(0)v_2 = \sum_{i \in V_1} \pi_i(0) - \sum_{i \in V_2} \pi_i(0)$, which is non-zero for any initial condition with unequal mass on the partition classes.

Let $\pi(0) = e_k^T$ denote a point-mass initial probability distribution localized at node $k \in V_1$. We evaluate the state vector after t steps:

$$\pi(t) = (\pi(0)v_1)(1)^t \pi^* + (\pi(0)v_2)(-1)^t w^T + \sum_{\alpha \geq 3} (\pi(0)v_\alpha) \mu_\alpha^t u_\alpha^T$$

Evaluating the projection coefficients yields $c_1 = e_k^T v_1 = 1$ and $c_2 = e_k^T v_2 = v_{2,k} = 1$. As $t \rightarrow \infty$, the decaying modes vanish, yielding the asymptotic equivalence:

$$\pi(t) \sim \pi^* + (-1)^t w^T$$

Taking the limit along alternating subsequences of discrete time steps reveals a permanent limit cycle:

$$\lim_{\substack{t \rightarrow \infty \\ t \text{ even}}} \pi_i(t) = \begin{cases} 2\psi_{0,i}^2 & \text{if } i \in V_1 \\ 0 & \text{if } i \in V_2 \end{cases}$$

$$\lim_{\substack{t \rightarrow \infty \\ t \text{ odd}}} \pi_i(t) = \begin{cases} 0 & \text{if } i \in V_1 \\ 2\psi_{0,i}^2 & \text{if } i \in V_2 \end{cases}$$

Therefore, the limit $\lim_{t \rightarrow \infty} \pi(t)$ does not exist. Initializing the walker at node $k \in V_2$ yields $c_2 = -1$, inducing an identical, phase-inverted limit cycle.

IV. Conclusion

The claim that discrete-time MERW relaxes to $\pi^* = \psi_0^2$ is invalid for bipartite topologies, including square, hypercubic, and ladder lattices. Simulations on these systems do not mix; instead, the distribution retains a period-two oscillation. To recover π^* , one must average over parity sectors, introduce a lazy walk step, or transition to continuous-time. A numerical illustration is provided in the Supplemental Material [2].

References

- [1] Z. Burda, J. Duda, J.-M. Luck, and B. Waclaw, Localization of the maximal entropy random walk, Phys. Rev. Lett. 102, 160602 (2009).
- [2] J. Duda, Maximal entropy random walk periodicity simulation, (2026).